#4086

MISSISSIPPI STATE UNIVERSITY
DEPARTMENT OF PSYCHOLOGY

Interaction Between Recency Effect and Strategy Preferences in Problem Solving

Xinyu Xie (xx86@msstate.edu), Jarrod Moss (jarrod.moss@msstate.edu)

Department of Psychology, Mississippi State University

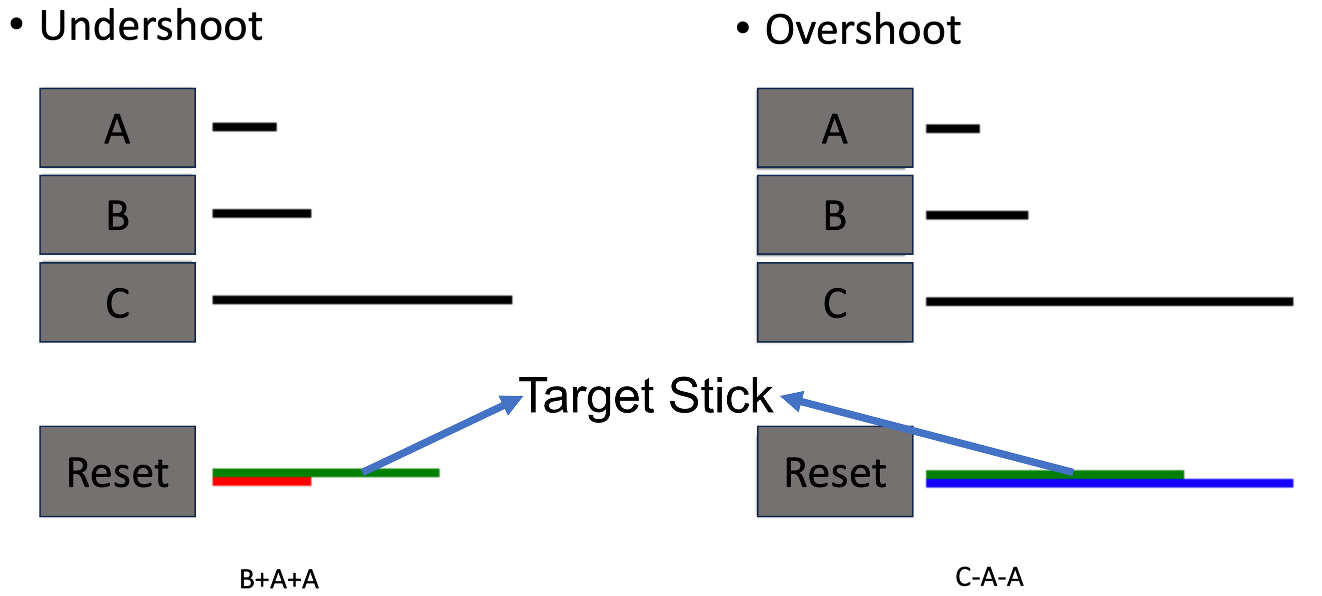
Problem Solving and Learning Lab

Introduction

- Do people tend to use the strategies they just used?
- Note that recent strategy use is not "experience" or "history", and it refers to one or several prior trials of using the same strategy before the current trial.
- The effect of recency of strategy use is also known as the perseveration effect, in which participants' strategy choices can be influenced by the strategy they used on the previous trial (Lemaire & Lecacheur, 2010; Scarampi & Gilbert, 2020; Schillemans et al., 2009).
- One explanation for this perseveration effect is that there is a strategy switch cost (Ho et al., 2023).
- However, this recency effect has not been explored extensively in prior research, so **this study aims to investigate the impact of recency of strategy use on strategy selection.**

Methods

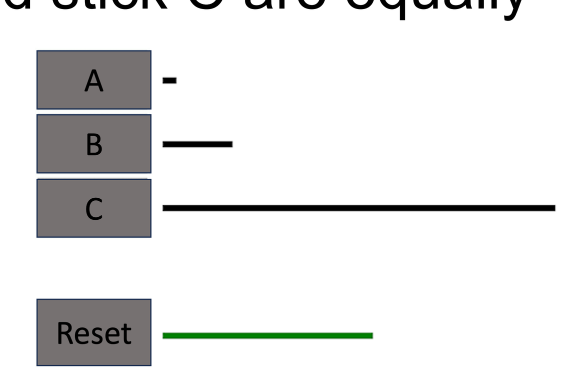
- **Participants**: N = 168
- **Task**: Building Stick Task (BST; Schunn et al., 2001)
  - The goal of the task is to add and subtract the lengths of three sticks to match the target length.
  - A participant's strategy can be categorized as either undershoot or overshoot based on the first move.



- **Procedure**:

Pretest → Strategy-learning → Posttest

- **Neutral** problems: stick B and stick C are equally close to the target stick
- Pre/Posttest phase:
  - 30 BST problems
  - Choose first stick
- Strategy-learning phase:
  - Another 30 BST problems
  - Solve every problem



- **Design**:

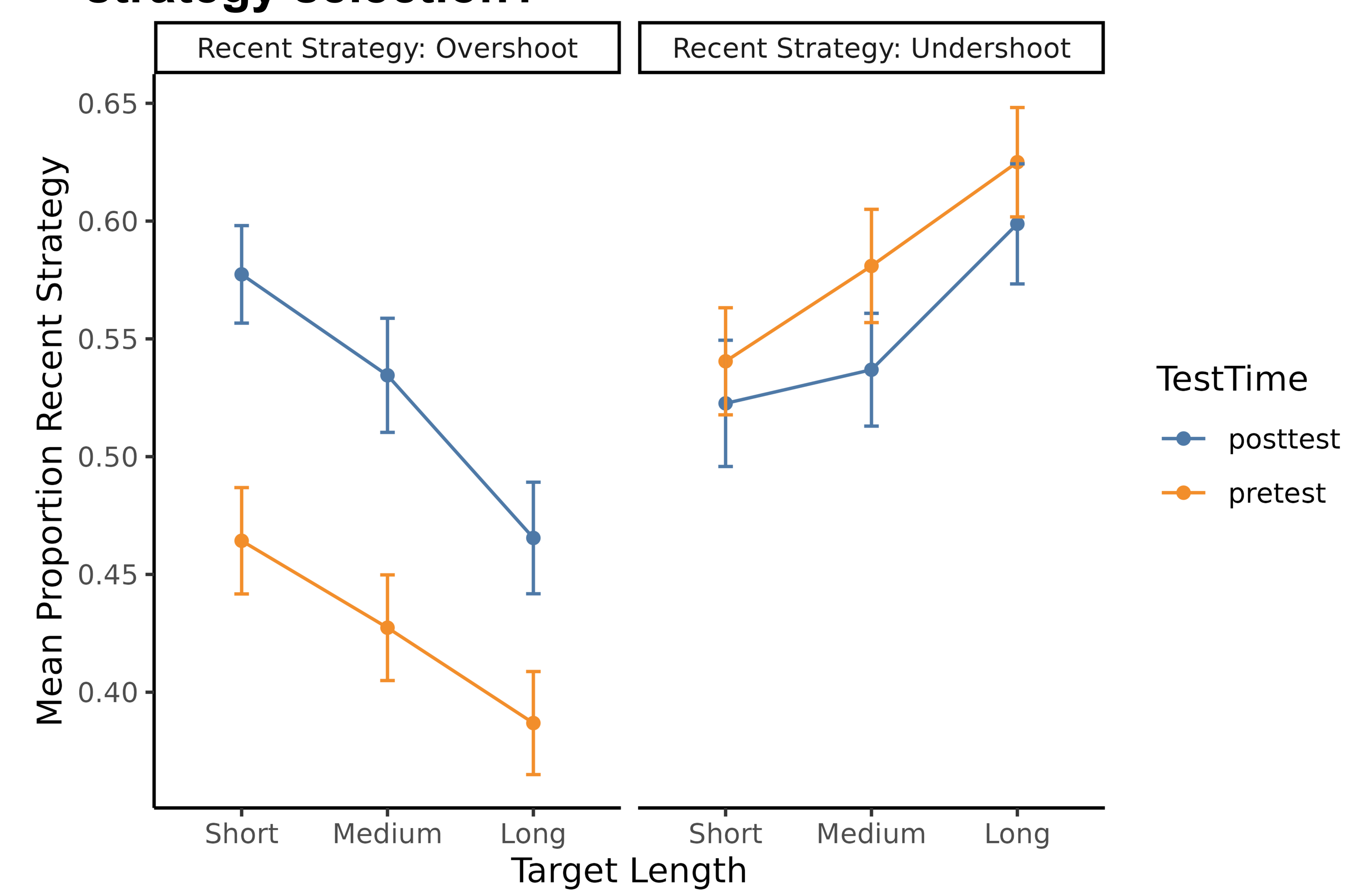
Pretest → Strategy-learning → Posttest

Condition 1	First 15 undershoot	Second 15 overshoot
Condition 2	First 15 overshoot	Second 15 undershoot

It only differed in terms of what **the recent strategy** is.

Results

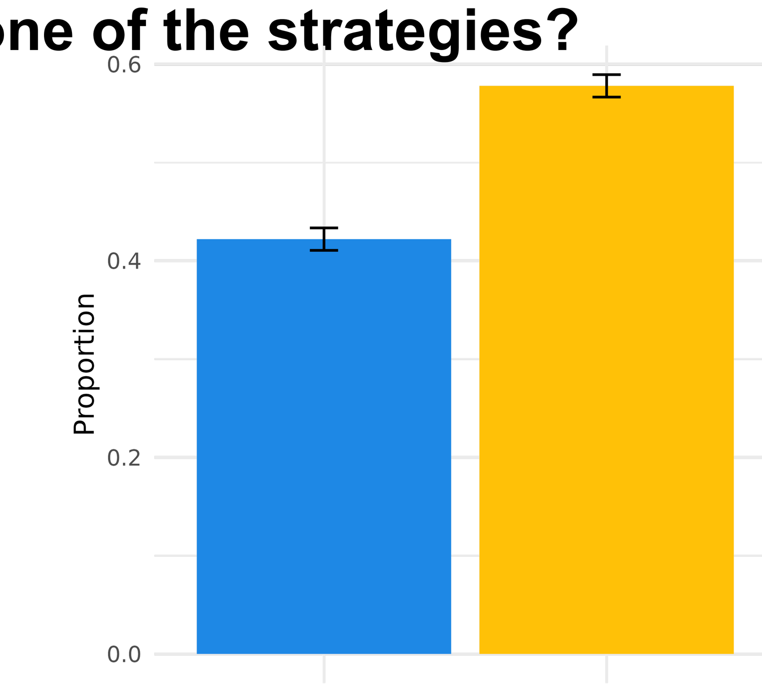
1. Does recent strategy use have an impact on strategy selection?



- Recency effect for overshoot, $b = -0.43$, $z = -4.60$, $p < .001$
- But not for undershoot, $b = 0.13$, $z = 1.37$, $p = .52$.

2. Do people prefer one of the strategies?

- Overshoot: 42% vs. Undershoot: 58%
- Compared to .5: $t(167) = 6.83$, $p < .001$ (BF > 1000)



3. Is there a difference between the two strategies?

- First response time: $t(165) = 1.19$, $p = .23$ (BF = 0.17)
- Solution time: $t(165) = 3.94$, $p < .001$ (BF = 129.44)
  - Overshoot: 3.6 (s) vs Undershoot: 3.25 (s)

Discussion

Summary:

- A recency effect was detected for the overshoot strategy, while such an effect was not observed for the undershoot strategy.
- People preferred the undershoot strategy more.
- Solution time for the undershoot strategy was less than that for the overshoot strategy while the first response time did not differ between the two strategies.

Discussion:

- Faster solution time is an explanation of why people prefer the undershoot strategy.
- Recency effect can be viewed as an outcome of avoiding a strategy switch cost (C1).
- Undershoot selection can be considered as an outcome of faster solution time, which can lead to an estimation of less execution time (C2).
- Consequently, the interaction between strategies and recency can be a comparison between C1 and C2.

Assuming the overshoot execution time is C, the undershoot execution time is C-C2.

When recent strategy is overshoot	Undershoot: C-C2+C1	C1 vs. C2
	Overshoot: C	
When recent strategy is undershoot	Undershoot: C-C2	
	Overshoot: C+C1	

Acknowledgement & References

This work was supported by Office of Naval Research grant N00014-21-1-2617.

Ho, M. K., Cohen, J. D., & Griffiths, T. (2023). *Rational simplification and rigidity in human planning. Psychological science, 9567976231200547. Advance online publication.*

Lemaire, P., & Lecacheur, M. (2010). *Strategy switch costs in arithmetic problem solving. Memory & Cognition, 38*(3), 322–332.

Lovett, M. C., & Schunn, C. D. (1999). *Task representations, strategy variability, and base-rate neglect. Journal of Experimental Psychology: General, 128*(2), 107–130.

Scarampi, C., & Gilbert, S. J. (2020). *The effect of recent reminder setting on subsequent strategy and performance in a prospective memory task. Memory, 28*(5), 677–691.

Schillemans, V., Luwel, K., Bulté, I., Onghena, P., & Verschaffel, L. (2009). *The Influence of Previous Strategy Use on Individuals' Subsequent Strategy Choice: Findings from a Numerosity Judgement Task. Psychologica Belgica, 49*, 49–4.